P20

Formerly S20

Ultra-Wide FoV RGBD Camera

MRDVS®

Empowering robots to see the world

V1.2 | 2026-09-15



Note: The image is a rendering of the enclosed version. The actual product may differ. A MIPI module version is also available.

The MRDVS P20 is a compact, cost-effective industrial RGBD camera with a near-panoramic field of view and long-range depth sensing. Built on dToF technology, it offers an ultra-wide horizontal FoV (≥170°) and 0.3-50 m ranging indoors and outdoors. RGB images and LiDAR point clouds are aligned in space and time. An integrated 200 Hz IMU supports mainstream SLAM algorithms, including the FastLIO family. P20 features an onboard SoC and is also available as a smaller MIPI module for easy integration into lawn mowers, UAVs, quadruped robots and humanoid robots for mapping and localization.

Key Features

dToF Depth Sensing

Calculates distance from the round-trip time of laser pulses, with a 0.2-50 m sensing range and ambient-light resistance up to 100 kLux.

Synchronized Data Output

Simultaneously outputs depth, high-quality RGB and IR images, with spatial and temporal synchronization.

Fully Solid-State Design

No moving mechanical parts. Compact and resistant to vibration, with aligned RGBD data and a MIPI module option.

Ultra-Wide FoV for SLAM

An ultra-wide horizontal FoV (≥170°) and a built-in high-rate IMU enhance SLAM stability in an integrated solid-state LiDAR and vision solution.

Applications



Robotic Lawn Mowers

Enables precise localization and navigation during autonomous operation in gardens, lawns and similar environments.



Industrial Quadruped Robots

Enhances localization, navigation and perception for research, inspection, exploration and complex industrial applications.



Humanoid Robots

Enables environmental perception for data collection, industrial tasks, home services, and patrol and localization in retail environments.



Multirotor UAVs

Improves localization, navigation, obstacle avoidance and environmental perception for inspection and surveying.

400-025-6680

Suite 7-902, China AI Town, Wenyi West Road, Yuhang District, Hangzhou, China

www.mrdvs.cn

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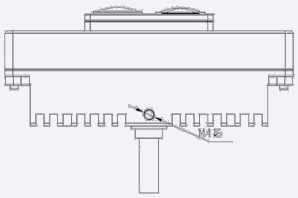
MRDVS®

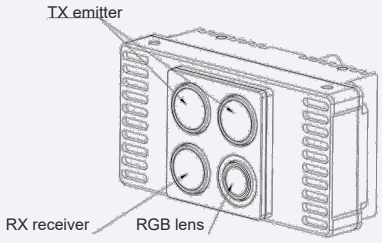
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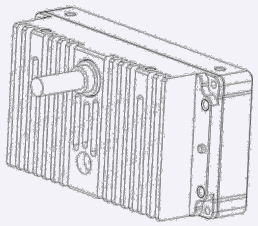
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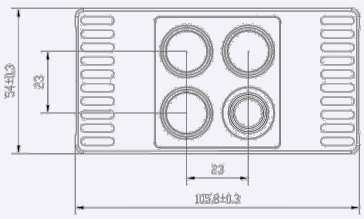
Product Specifications

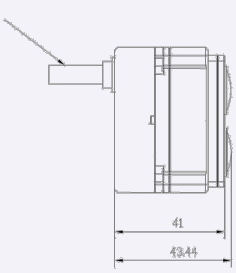
Product model	P20
Operating principle	dToF
Output formats	Depth / IR / point cloud / RGB
ToF resolution and frame rate	240 × 96 @ 10 fps / 240 × 92 @ 10 fps (VW version)
ToF field of view (H × V)	170° × 65° / 170° × 130° (VW version)
RGB resolution and frame rate	1364 × 1124 @ 10 fps
RGB field of view (H × V)	170° × 140°
Measuring range	0.2-50 m / 0.2-35 m (VW version) At 10% reflectivity and 100 kLux
Accuracy	≤ ±30 mm @ 20 m; ±50 mm @ 50 m (1σ)
Average operating power	< 9 W
Peak power	< 20 W
Laser wavelength	940 nm
Dimensions	105 × 54 × 44 mm Current version; final dimensions may change.
Time synchronization	gPTP
Weight	Approx. 420 g (actual product weight prevails)
Supply voltage	12-27 V
Communication interfaces	Ethernet / MIPI (customization required) / M12 circular connector
Ingress protection	IP67 (enclosed unit)
Operating temperature	-20°C to +75°C
Storage temperature	-40°C to +85°C
Ambient-light resistance	100 kLux

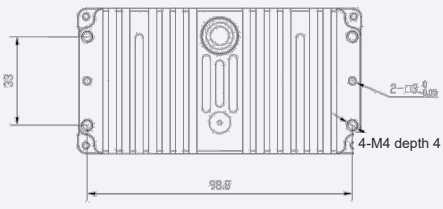


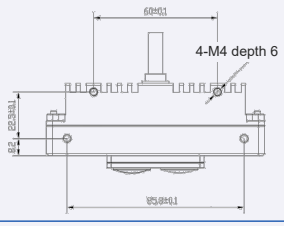












Note: Dimensions shown are for the current version.
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